

HTB SHERLOCK FORTYSEVEN-1

Forensic-Quality Walkthrough

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Quick Reference: All 15 Answers

#	Question	Answer
1	Primary name of the APT group in the SecureList report?	Mysterious Elephant
2	Since which year has this group's attack activity been dated back to?	2022
3	First malicious exported entry function of ORPCBackdoor?	GetFileVersionInfoByHandleEx (void)
4	File checked by the backdoor before creating persistence?	ts.dat
5	Other well-known South Asian APT group linked via the backdoor?	BITTER
6	Backdoor consistently used since 2023, C2 evolved from TCP to HTTPS?	AsyncShell
7	Minimum number of processes for MemLoader HidenDesk to proceed?	40
8	Name of the hidden desktop created by MemLoader HidenDesk?	MalwareTech_Hidden
9	MITRE ATT&CK ID for 'Registry Run Keys / Startup Folder'?	T1547.001
10	Tool that recursively searches Desktop and Downloads folders?	Stom Exfiltrator
11	MITRE ATT&CK ID for the 'PowerShell' technique?	T1059.001
12	Downloader previously associated with Origami Elephant?	Vtyrei

#	Question	Answer
13	CVE exploited in the January 2024 Asyncshell archive campaign?	CVE-2023-38831
14	MD5 hash of ChromeStealer Exfiltrator sample WhatsAppOB.exe?	9e50adb6107067ff0bab73307f5499b6
15	MITRE ATT&CK ID for 'Exfiltration Over C2 Channel'?	T1041

Detailed Task Walkthroughs

Each task includes the evidence source, key quote, final answer, and timestamped screenshot.

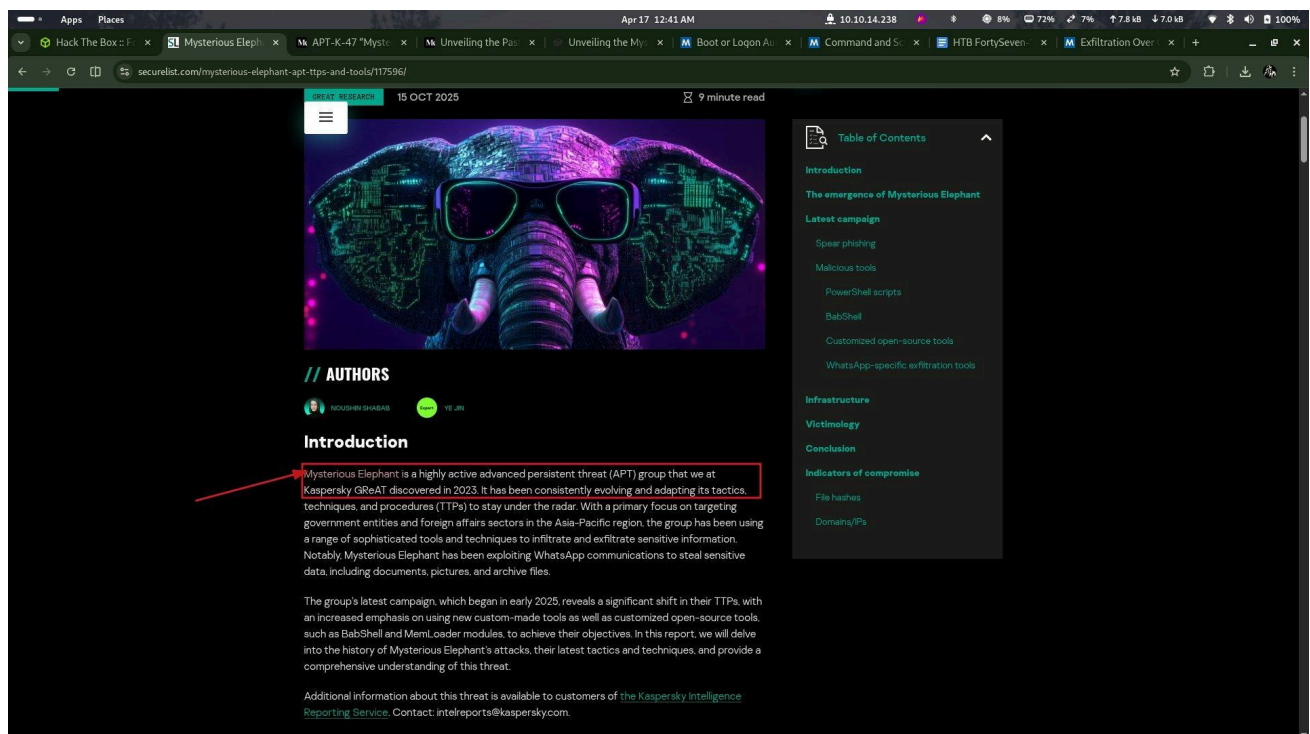
Task 1: Primary name of the APT group in the SecureList report?

Evidence -1 (Kaspersky SecureList) :

Mysterious Elephant is a highly active advanced persistent threat (APT) group...

Ans=> Mysterious Elephant

Screenshot 1: Evidence-1 Introduction section, 'Mysterious Elephant' highlighted



Task 2: Since which year has this group's attack activity been dated back to?

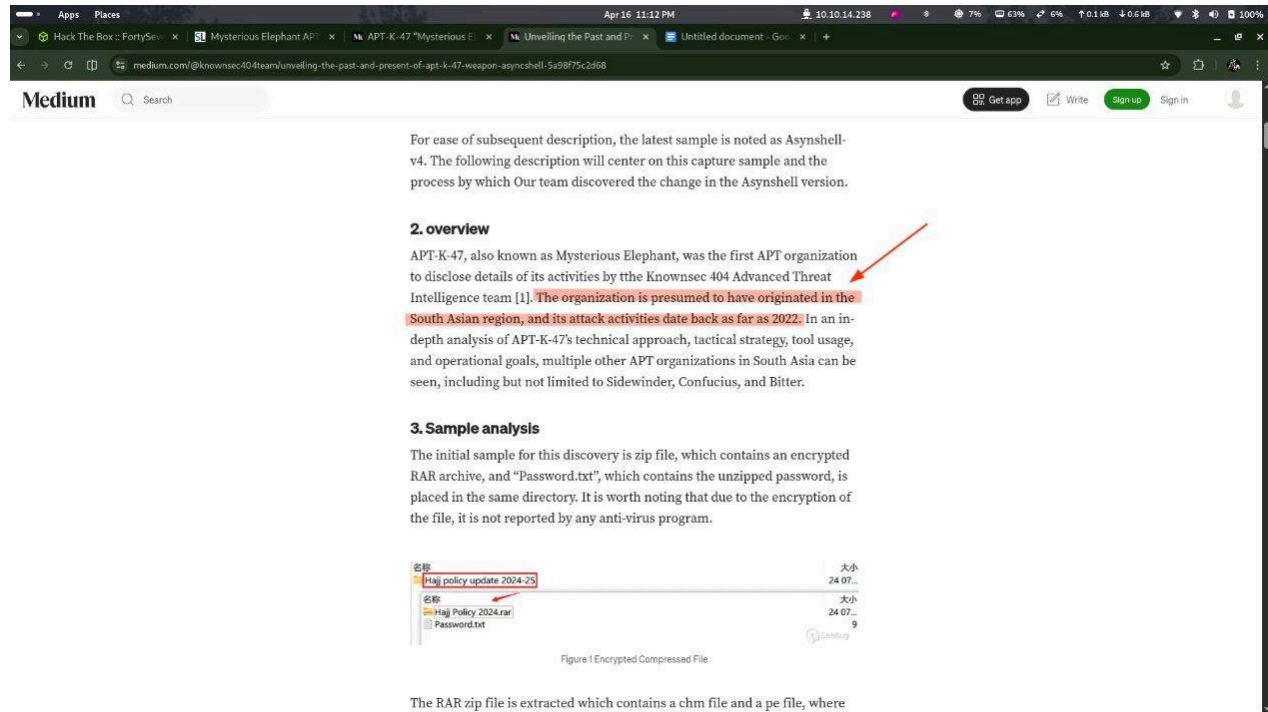
Evidence Source: Evidence

Evidence -3 (Medium / Knownsec):

The organization is presumed to have originated in the South Asian region, and its attack activities date back as far as 2022.

ANS => 2022

Screenshot 2: Evidence-3 – Section 2 Overview, '2022' highlighted



Task 3: First malicious exported entry function of ORPCBackdoor?

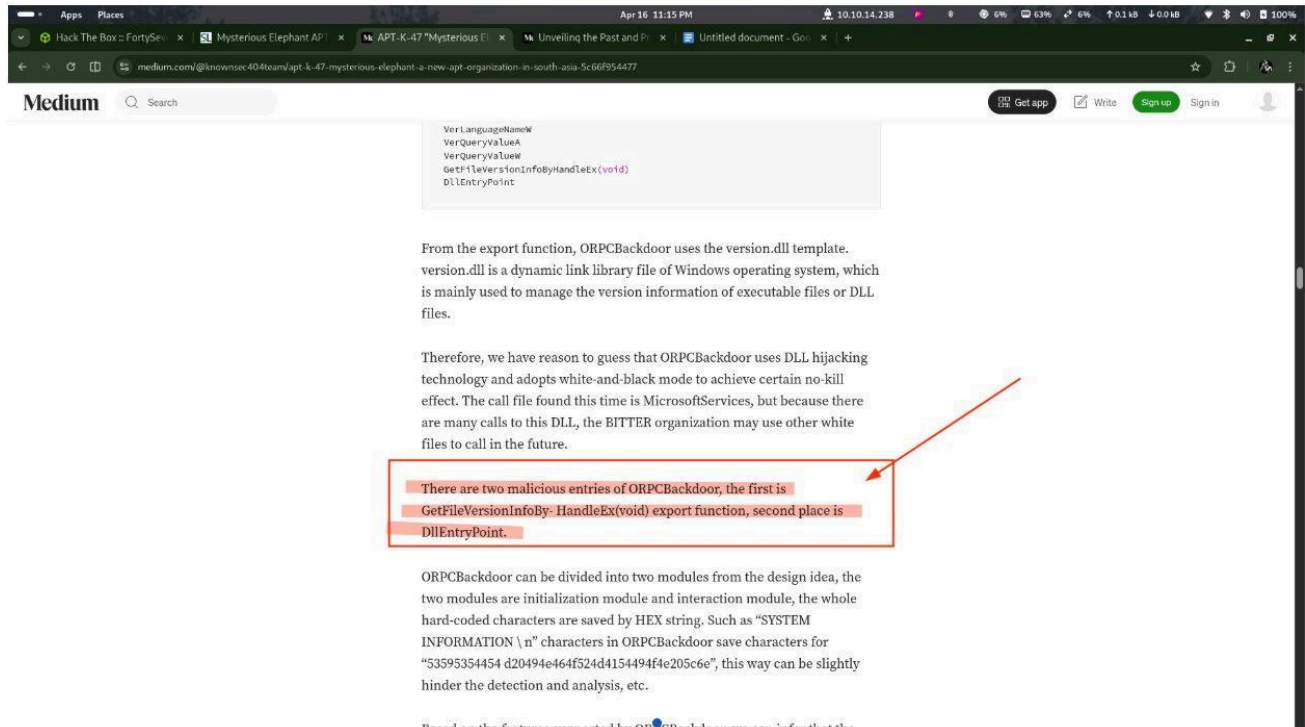
Evidence Source:

Evidence 2 (Medium / Knownsec):

There are two malicious entries of ORPCBackdoor the first is `GetFileVersionInfoByHandleEx(void)` export function, second place is `DllEntryPoint`.

Ans => `GetFileVersionInfoByHandleEx(void)`

Screenshot 3: Evidence-2 Export function section, '`GetFileVersionInfoByHandleEx(void)`' highlighted



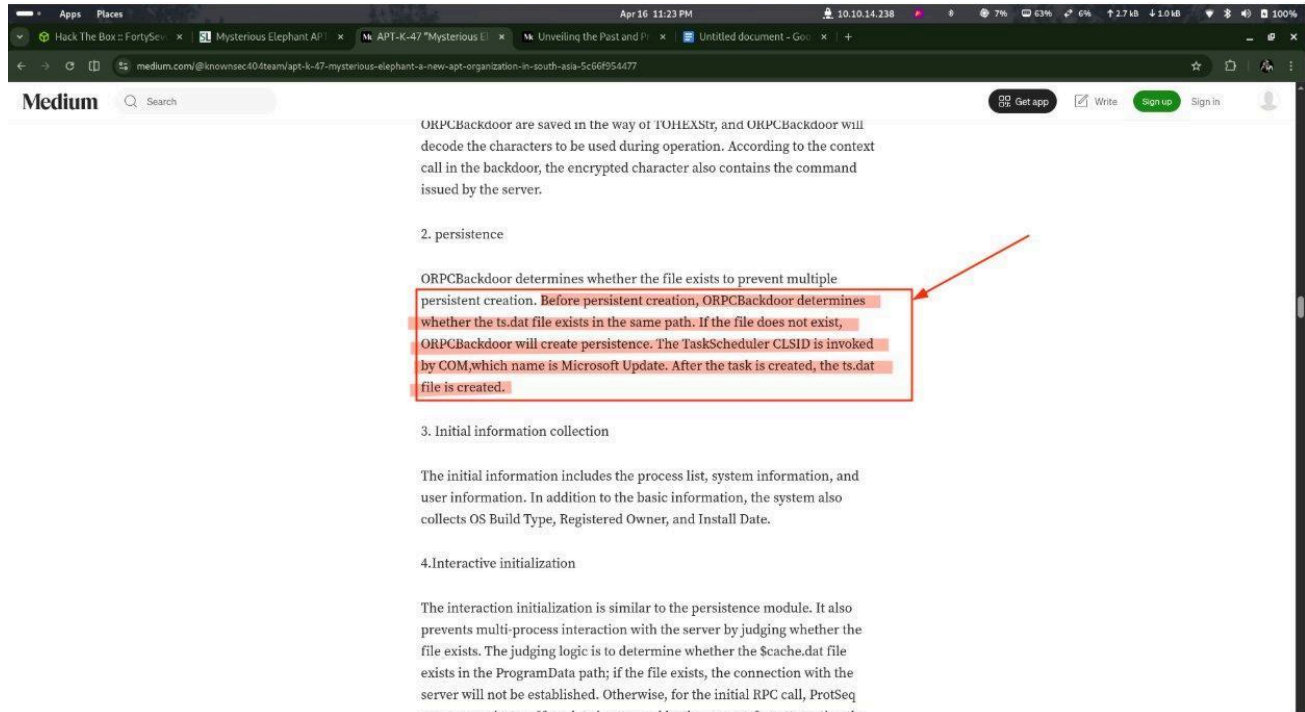
Task 4: File checked by the backdoor before creating persistence?

Evidence-2 (Medium / Knownsec) :

Before persistent creation, ORPCBackdoor determines whether the ts.dat file exists in the same path. If the file does not exist, ORPCBackdoor will create persistence."

ANS=> ts.dat

Screenshot 4: Evidence-2 Section 2 Persistence, 'ts.dat' highlighted



Task 5: Other well-known South Asian APT group linked via the backdoor?

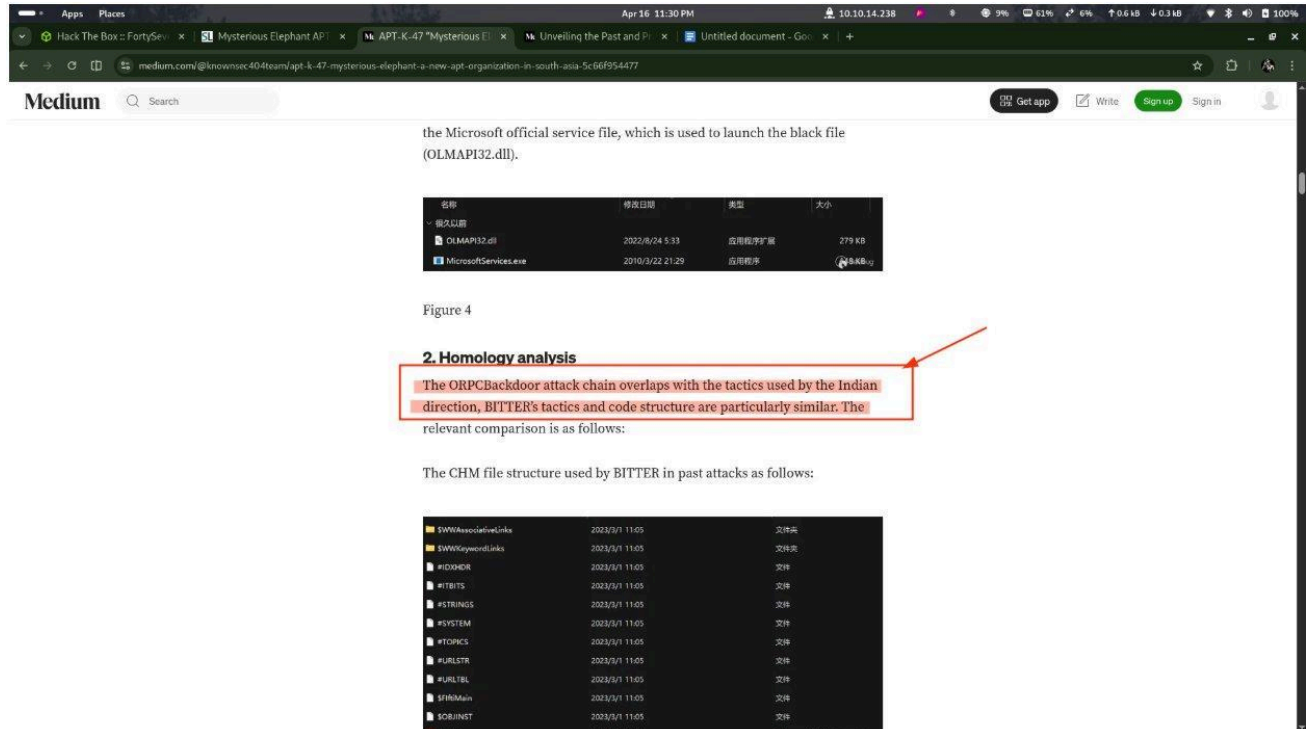
Evidence Source: Evidence

Evidence-2 (Medium / Knownsec):

The ORPCBackdoor attack chain overlaps with the tactics used by the Indian direction, BITTER's tactics and code structure are particularly similar

ANS => Bitter`s

Screenshot 5: Evidence-2 Homology Analysis section, 'BITTER' highlighted



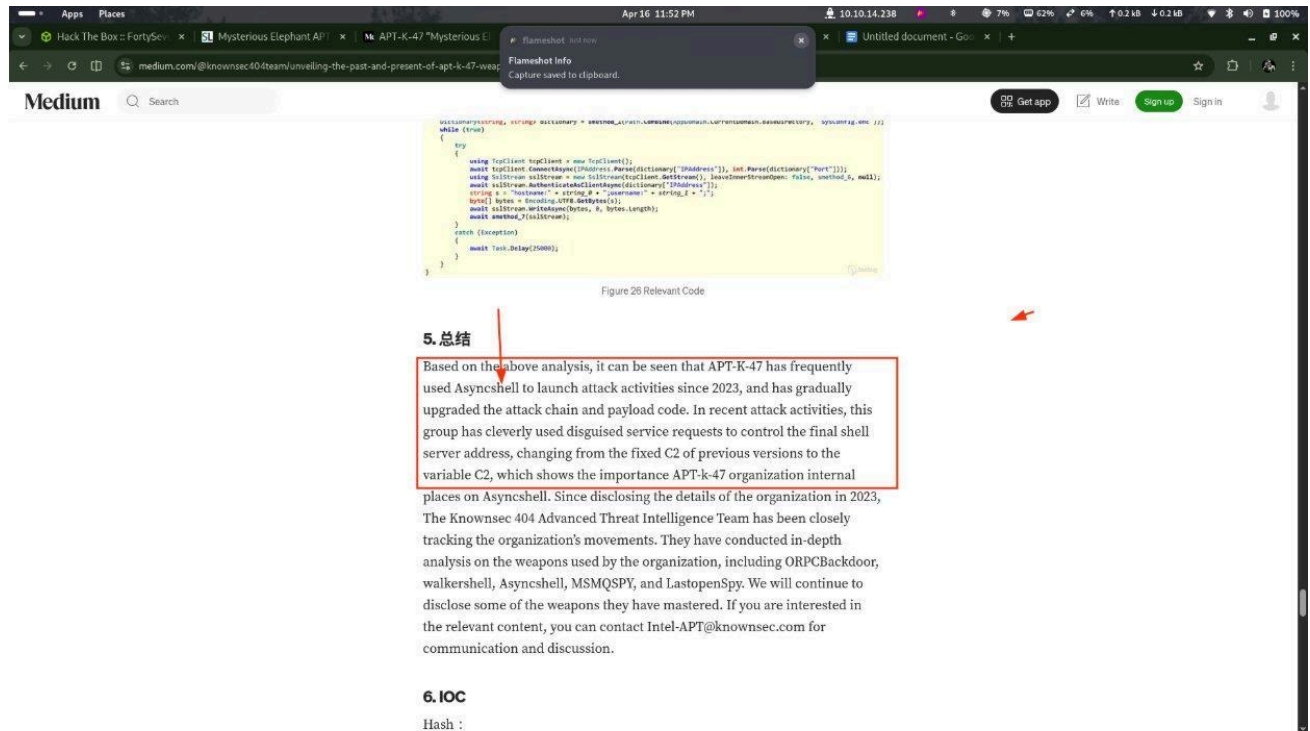
Task 6: Backdoor consistently used since 2023, C2 evolved from TCP to HTTPS?

Evidence -3 (Medium / Knownsec) :

APT-K-47 has frequently used Asyncshell to launch attack activities since 2023, and has gradually upgraded the attack chain and payload code... changing from the fixed C2 of previous versions to the variable C2.

Ans => AsyncShell

Screenshot 6: Evidence-3 Section 5 Summary, Asyncshell since 2023 + C2 evolution highlighted



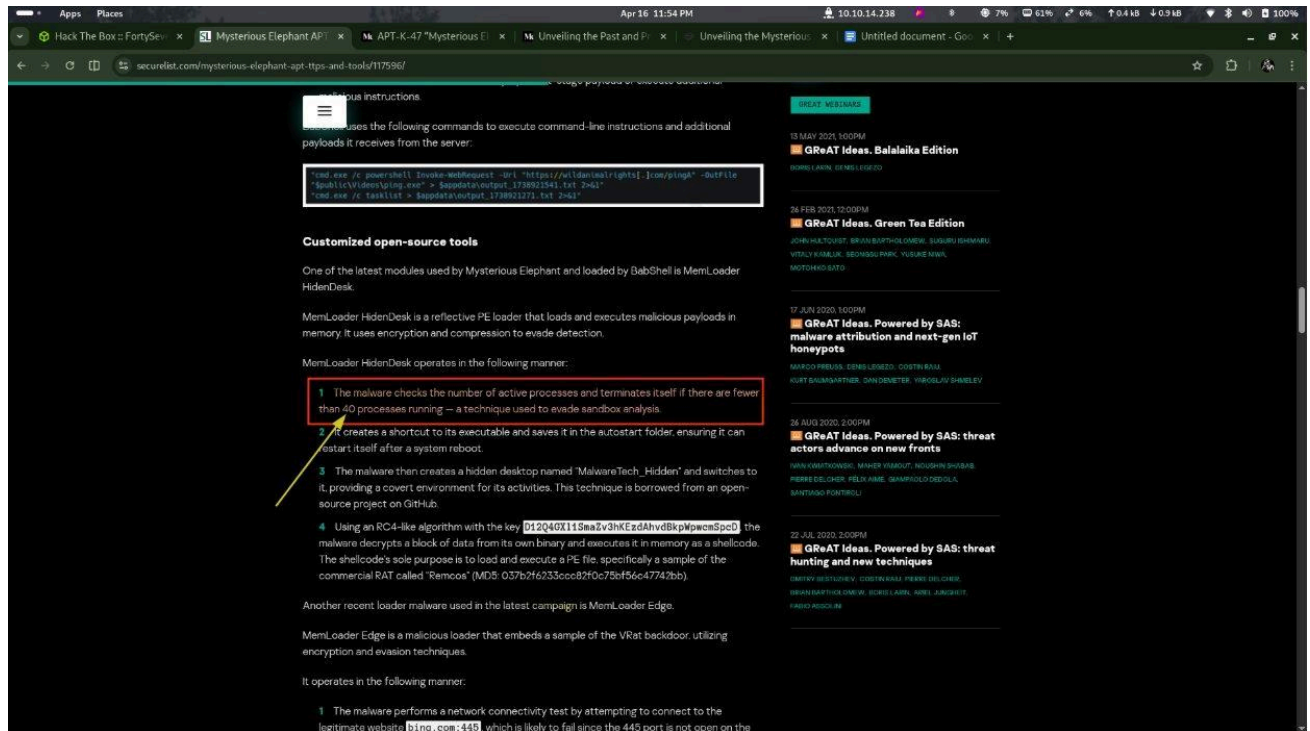
Task 7: Minimum number of processes for MemLoader HidenDesk to proceed?

Evidence1 (Kaspersky SecureList)

terminates itself if there are fewer than 40 processes running

ANS => 40

Screenshot 7: Evidence-1 MemLoader HidenDesk section, '40 processes' highlighted



Task 8: Name of the hidden desktop created by MemLoader HidenDesk?

Evidence 1 (Kaspersky SecureList)

creates a hidden desktop named `MalwareTech_Hidden` and switches to it

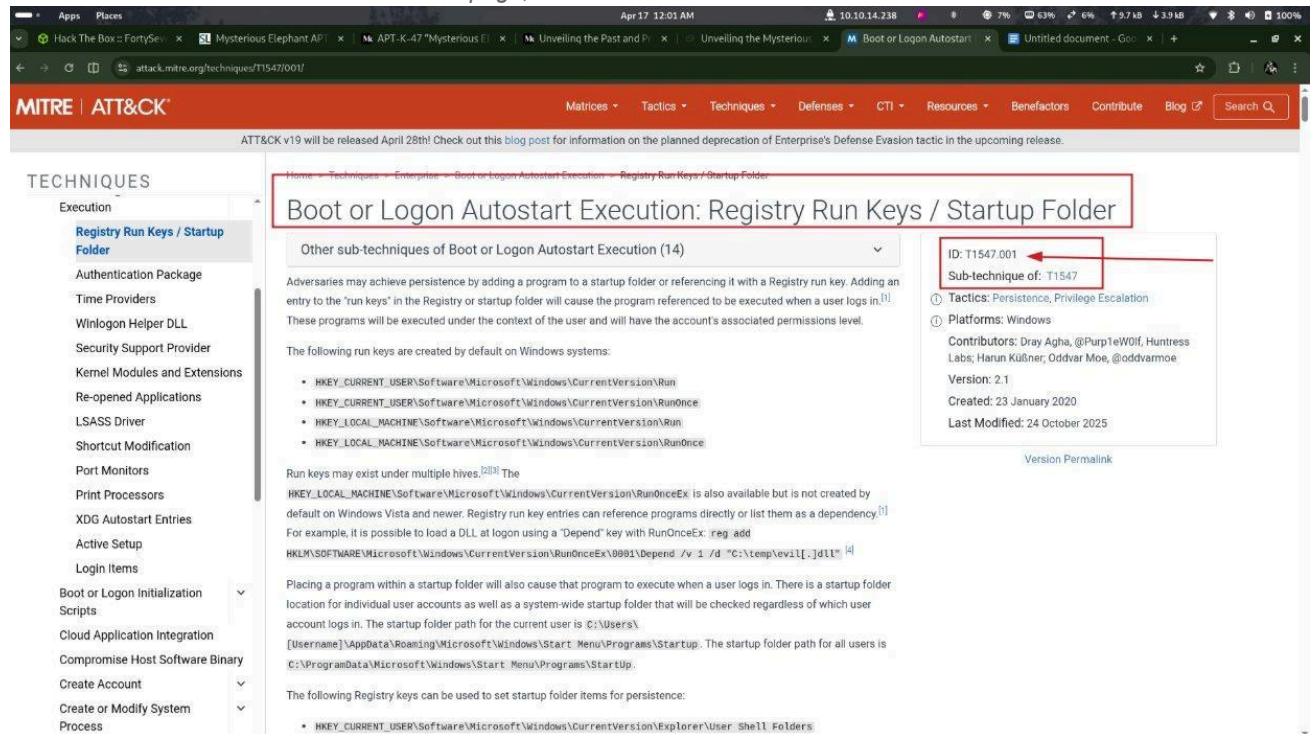
Ans => `MalwareTech_Hidden`

Screenshot 8: Evidence-1 MemLoader HidenDesk section, 'MalwareTech_Hidden' highlighted

Evidence MITRE ATT&CK Framework :

ANS => T1547.001

Screenshot 9: MITRE ATT&CK T1547.001 page, ID



Task 10: Tool that recursively searches Desktop and Downloads folders?

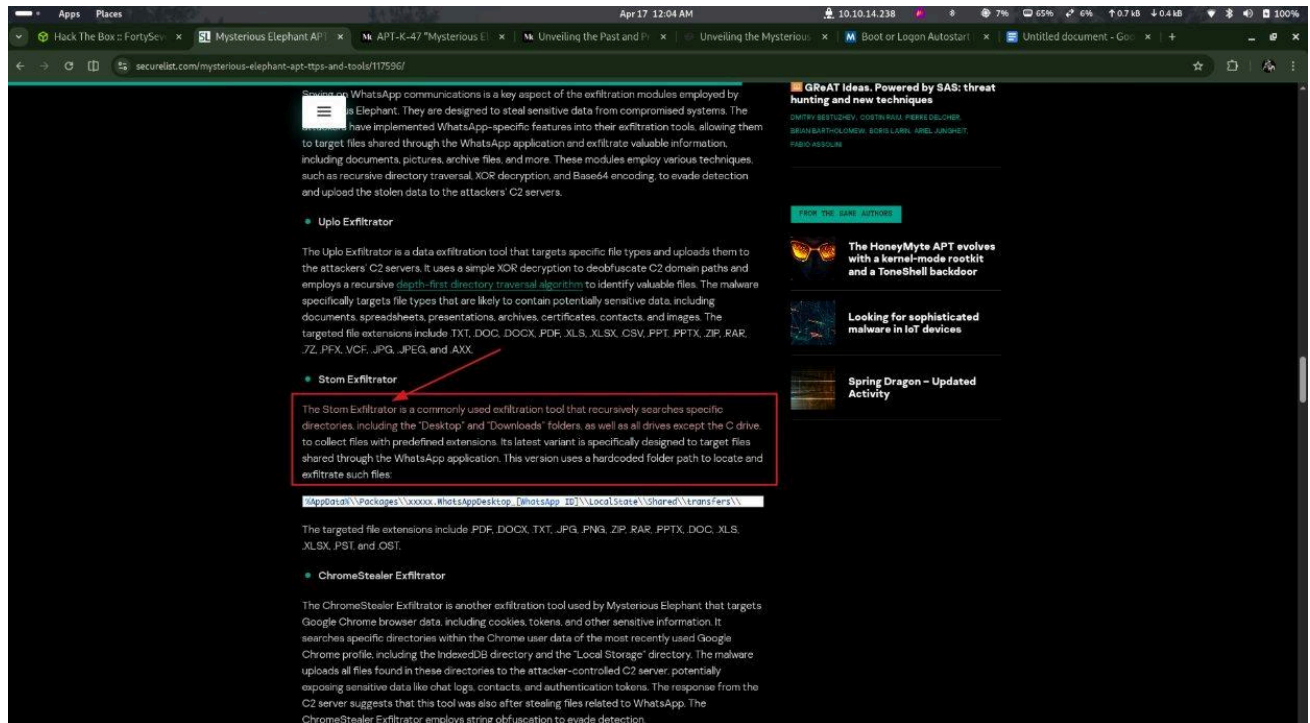
Evidence Source: Evidence-1 (Kaspersky SecureList)

Evidence -1 (Kaspersky SecureList):

Stom Exfiltrator.. recursively searches specific directories, including the Desktop and Downloads folders

Ans => Stom Exfiltrator

Screenshot 10: Evidence-1 Exfiltration section, 'Stom Exfiltrator' highlighted



Task 11: MITRE ATT&CK ID for the 'PowerShell' technique?

Evidence MITRE ATT&CK Framework:

Command and Scripting Interpreter: PowerShell ID: T1059.001

Ans => T1059.001

Screenshot 11: MITRE ATT&CK T1059.001 page, ID highlighted

MITRE | ATT&CK

Home > Techniques > Enterprise > Command and Scripting Interpreter > PowerShell

Command and Scripting Interpreter: PowerShell

Other sub-techniques of Command and Scripting Interpreter (13)

Adversaries may abuse PowerShell commands and scripts for execution. PowerShell is a powerful interactive command-line interface and scripting environment included in the Windows operating system.^[1] Adversaries can use PowerShell to perform a number of actions, including discovery of information and execution of code. Examples include the `Start-Process` cmdlet, which can be used to run an executable and the `Invoke-Command` cmdlet which runs a command locally or on a remote computer (though administrator permissions are required to use PowerShell to connect to remote systems).

PowerShell may also be used to download and run executables from the Internet, which can be executed from disk or in memory without touching disk.

A number of PowerShell-based offensive testing tools are available, including [Empire](#), [PowerSploit](#), [PoshC2](#), and [PSAttack](#).^[2]

PowerShell commands/scripts can also be executed without directly invoking the `powershell.exe` binary through interfaces to PowerShell's underlying `System.Management.Automation` assembly DLL exposed through the .NET framework and Windows Common Language Interface (CLI).^{[3][4]}

Procedure Examples

ID	Name	Description
C0025	2016 Ukraine Electric Power Attack	During the 2016 Ukraine Electric Power Attack, Sandworm Team used PowerShell scripts to run a credential harvesting tool in memory to evade defenses. ^[5]
C0034	2022 Ukraine Electric Power Attack	During the 2022 Ukraine Electric Power Attack, Sandworm Team utilized a PowerShell utility called TANKTRAP to spread and launch a wiper using Windows Group Policy. ^[7]
S0677	AADInternals	AADInternals is written and executed via PowerShell. ^[8]

Version Permalink

Task 12: Downloader previously associated with Origami Elephant?

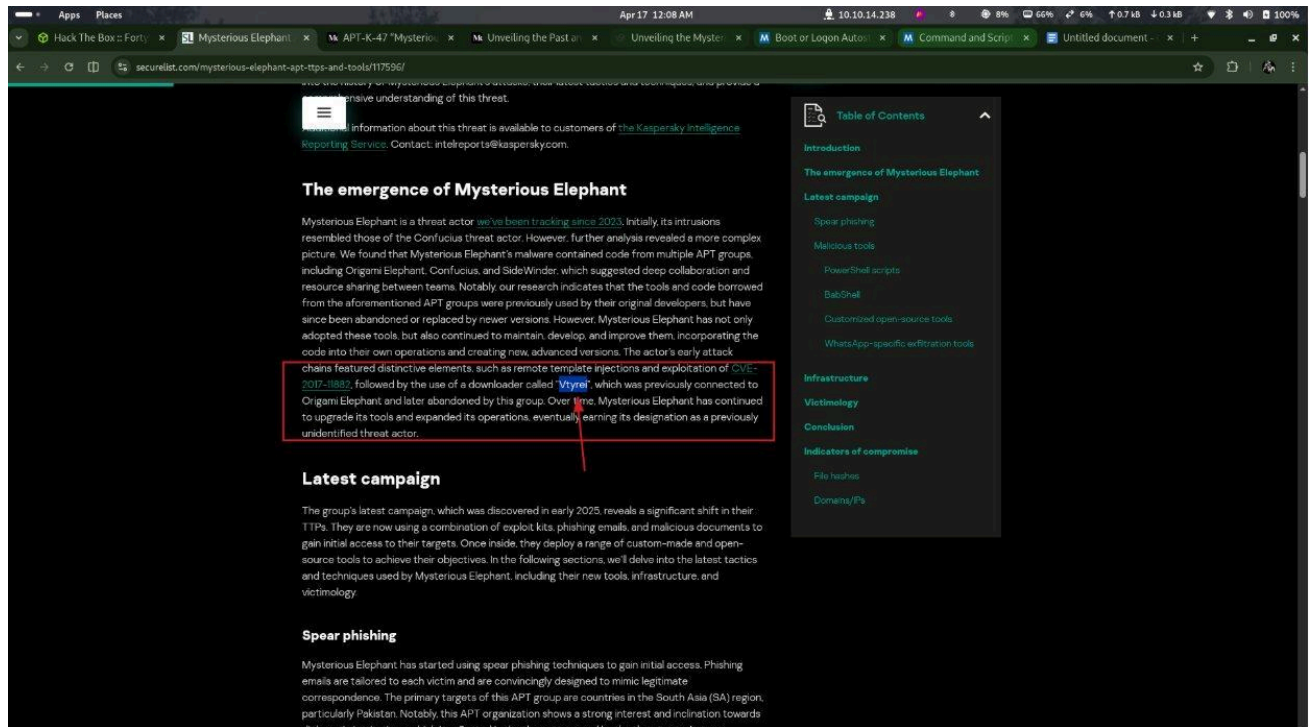
Evidence Source: Evidence-1 (Kaspersky SecureList)

Evidence -1 (Kaspersky SecureList):

a downloader called Vtyrei, which was previously connected to Origami Elephant

Ans => Vtyrei

Screenshot 12: Evidence-1 Emergence section, 'Vtyrei' + 'Origami Elephant' highlighted



Task 13: CVE exploited in the January 2024 Asyncshell archive campaign?

Evidence3: malicious sample exploiting the CVE-2023-38831 vulnerability

Ans => CVE-2023-38831

Screenshot 13: Evidence-3 Section 4.1, CVE-2023-38831 highlighted

Medium

版本	发布日期	传输协议	是否加密	漏洞/功能	加密方式	执行环境
v1	2023.9	TCP	√	CVE-2023-38831 CHM加载执行	硬编码	CMD Powershell
v2	2024.4	Https	√	CHM加载执行 PDF文档伪装	硬编码	CMD
v3	2024.7	TCP	√	Ink+vbs	解密文件	CMD
v4	2024.9	Http TCP	×	CHM加载执行	服务端分发	CMD (Powercat)

The following describes the discovery process and version updates of Asyncshell on a timeline.

4.1 Discover Asyncshell for the first time

Our team first discovered Asyncshell back in January 2024, when we found a malicious sample exploiting the **CVE-2023-38831** vulnerability, with the overall attack chain shown below:

Figure 6 Attack Chain

Attackers used content related to the remuneration of civil servants and temporary civil servants on temporary leave in the field as bait:

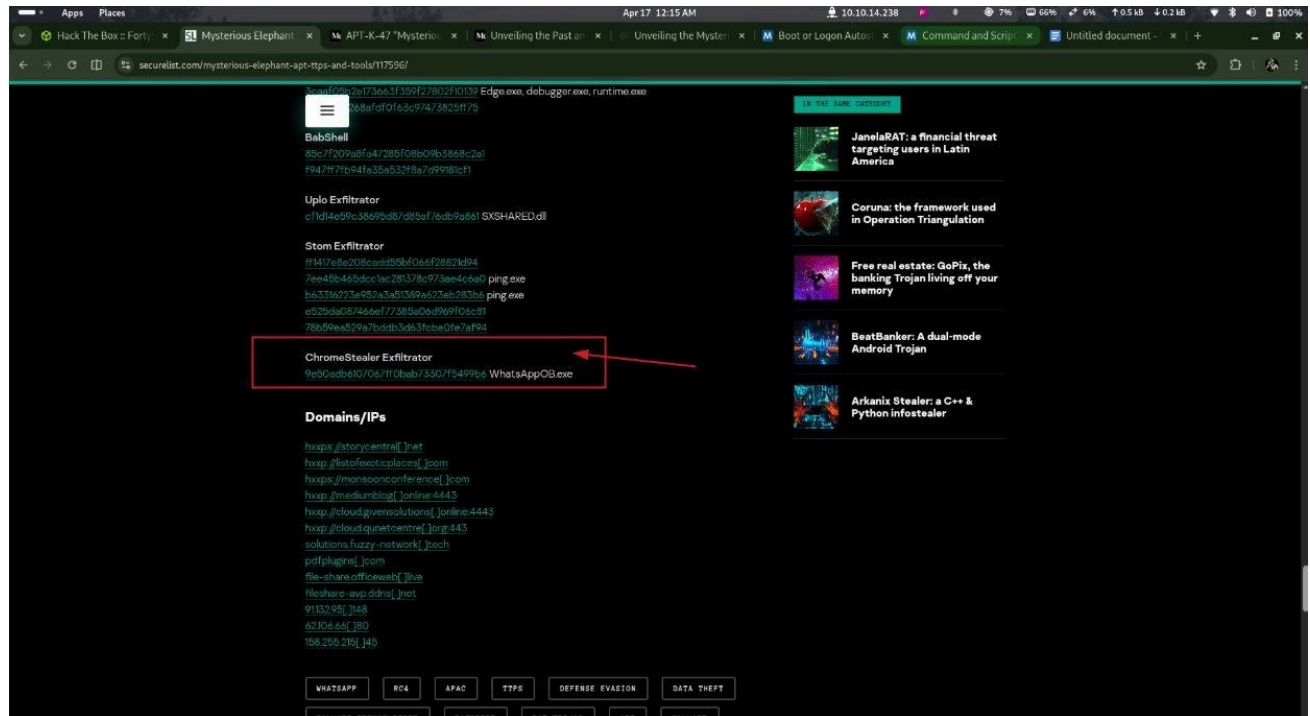
Task 14: MD5 hash of ChromeStealer Exfiltrator sample WhatsAppOB.exe?

Evidence-1 IOC Table (Kaspersky):

ChromeStealer Exfiltrator 9e50adb6107067ff0bab73307f5499b6 WhatsAppOB.exe

Ans => 9e50adb6107067ff0bab73307f5499b6

Screenshot 14: Evidence-1 IOC table, ChromeStealer MD5 + WhatsAppOB.exe highlighted



Task 15: MITRE ATT&CK ID for 'Exfiltration Over C2 Channel'?

Evidence MITRE ATT&CK Framework: *Exfiltration Over C2 Channel* ID: T1041

Ans => T1041

Screenshot 15: MITRE ATT&CK T1041 page, ID highlighted

MITRE | ATT&CK

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ATT&CK v19 will be released April 28th! Check out this blog post for information on the planned deprecation of Enterprise's Defense Evasion tactic in the upcoming release.

TECHNIQUES

PersistencePrivilege EscalationDefense EvasionCredential AccessDiscoveryLateral MovementCollectionCommand and ControlExfiltrationAutomated ExfiltrationData Transfer Size LimitsExfiltration Over Alternative ProtocolExfiltration Over C2 ChannelExfiltration Over Other Network MediumExfiltration Over Physical MediumExfiltration Over Web ServiceScheduled TransferTransfer Data to Cloud AccountImpactMobile

Home > Techniques > Enterprise > Exfiltration Over C2 Channel

Exfiltration Over C2 Channel

Adversaries may steal data by exfiltrating it over an existing command and control channel. Stolen data is encoded into the normal communications channel using the same protocol as command and control communications.

ID: T1041

Sub-techniques: No sub-techniques

Tactic: Exfiltration

Platforms: ESXi, Linux, Windows, macOS

Contributors: William Cain

Version: 2.3

Created: 31 May 2017

Last Modified: 24 October 2025

Version Permalink

Procedure Examples

ID	Name	Description
S0045	ADVSTORESHELL	ADVSTORESHELL exfiltrates data over the same channel used for C2. ^[1]
G1030	Agrius	Agrius exfiltrated staged data using tools such as Putty and WinSCP, communicating with command and control servers. ^[2]
S1025	Amadey	Amadey has sent victim data to its C2 servers. ^[3]
S0584	AppleJeuS	AppleJeuS has exfiltrated collected host information to a C2 server. ^[4]
S0622	AppleSeed	AppleSeed can exfiltrate files via the C2 channel. ^[5]
G0022	APT3	APT3 has a tool that exfiltrates data over the C2 channel. ^[6]

— End of Walkthrough —
All 15 tasks completed with evidence.